THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET A

Chapter 14: **Probability** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The probability of an impossible event is: 1

(a) 0 **(b)** 1 **(c)** $\frac{1}{2}$ **(d)** −1

2. If $P(E) = 0.35$, then $P(\text{not } E)$ is: 1

(a) 0.65 **(b)** 0.35 **(c)** 1.35 **(d)** 0.53

3. A die is thrown once. The probability of getting a number greater than 4 is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{6}$ **(d)** $\frac{2}{3}$

4. A card is drawn from a well-shuffled deck of 52 cards. The probability that it is an ace is: 1

(a) $\frac{1}{13}$ **(b)** $\frac{1}{52}$ **(c)** $\frac{4}{13}$ **(d)** $\frac{1}{4}$

5. A student states: "The probability of an event can be -0.2." The correct response is: 1

(a) Correct **(b)** Incorrect — $0 \le P(E) \le 1$

(c) Correct for impossible events **(d)** Cannot be determined

6. Two coins are tossed together. The probability of getting no head is: 1

(a) $\frac{1}{4}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{3}{4}$ **(d)** 0

7. A die is thrown. Student A says P(even prime) $= \frac{1}{6}$; Student B says P(even prime) $= \frac{1}{2}$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

8. Which of the following cannot be the probability of an event? 1

(a) $\frac{2}{3}$ **(b)** 0.7 **(c)** 115% **(d)** 0

9. A letter is chosen at random from the word ASSASSINATION. The probability that it is a vowel is: 1

(a) $\frac{6}{13}$ **(b)** $\frac{7}{13}$ **(c)** $\frac{5}{13}$ **(d)** $\frac{4}{13}$

10. The sum of the probabilities of all the elementary events of an experiment is: 1

(a) 0 **(b)** 1 **(c)** $\frac{1}{2}$ **(d)** It varies

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A box contains cards numbered 1 to 20. A card is drawn at random. Find the probability that the number is 2

(i) prime

(ii) a multiple of 5.

12. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting 2

(i) a king of red colour

(ii) a face card.

13. A student says: "The probability of a head in a coin toss is $\frac{1}{2}$, so in 10 tosses exactly 5 heads will come." Justify or refute. 2

14. A piggy bank contains hundred 50 p coins, fifty ₹1 coins, twenty ₹2 coins and ten ₹5 coins. One coin falls out at random. Find the probability that it is 2

(i) a 50 p coin

(ii) not a ₹5 coin.

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two dice are thrown at the same time. Find the probability of getting **3**
(i) a sum of 8
(ii) a doublet
(iii) a sum less than or equal to 4.

16. Tickets numbered 1 to 50 are mixed and one ticket is drawn at random. Find the probability that the number is **3**
(i) a multiple of 7
(ii) a perfect square
(iii) a multiple of 3 or 4.

17. 12 defective pens are accidentally mixed with 132 good ones. **3**
(i) One pen is taken out at random. Find the probability that it is good.
(ii) If a good pen is removed first and not replaced, find the probability that the next pen drawn is defective.

OR

A bag contains 5 red balls and some blue balls. If the probability of drawing a blue ball is double that of a red ball, find the number of blue balls. **3**

18. Two customers Shyam and Ekta visit a shop in the same week (Tuesday to Saturday). Each is equally likely to visit on any day. Find the probability that they visit on **3**
(i) the same day
(ii) consecutive days
(iii) different days.

OR

A jar contains 24 marbles, some green and others blue. The probability of drawing a green marble is $\frac{2}{3}$. **3**
(i) Find the number of blue marbles.
(ii) If 4 more blue marbles are added, find the new probability of a green marble.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** In a game of chance, an arrow is spun and comes to rest pointing at one of the numbers 1, 2, 3, …, 8, all equally likely. Find the probability that it points at **5**
(i) 8
(ii) an odd number
(iii) a number greater than 2
(iv) a number less than 9.
(v) A student says P(9) $= \frac{1}{8}$. Comment.

OR

All the black face cards are removed from a pack of 52 cards. A card is drawn at random from the remaining cards. Find the probability of getting **5**
(i) a face card
(ii) a red card
(iii) a black card
(iv) a king.

20. A student argues: "When two dice are thrown, the possible sums are 2, 3, …, 12 — eleven outcomes. So each sum has probability $\frac{1}{11}$." **5**
(i) How many outcomes are there in all?
(ii) Find P(sum = 2) and P(sum = 7).
(iii) Is the argument correct?
(iv) Find the probability that the sum is a prime number.

OR

A bag contains 12 balls of which x are white. **5**
(i) If one ball is drawn at random, what is the probability that it is white?
(ii) If 6 more white balls are put in the bag, the probability of drawing a white ball becomes double the earlier probability. Find x.
(iii) Find the probability of not drawing a white ball after the 6 balls are added.

END OF QUESTION PAPER — Total: 40 Marks